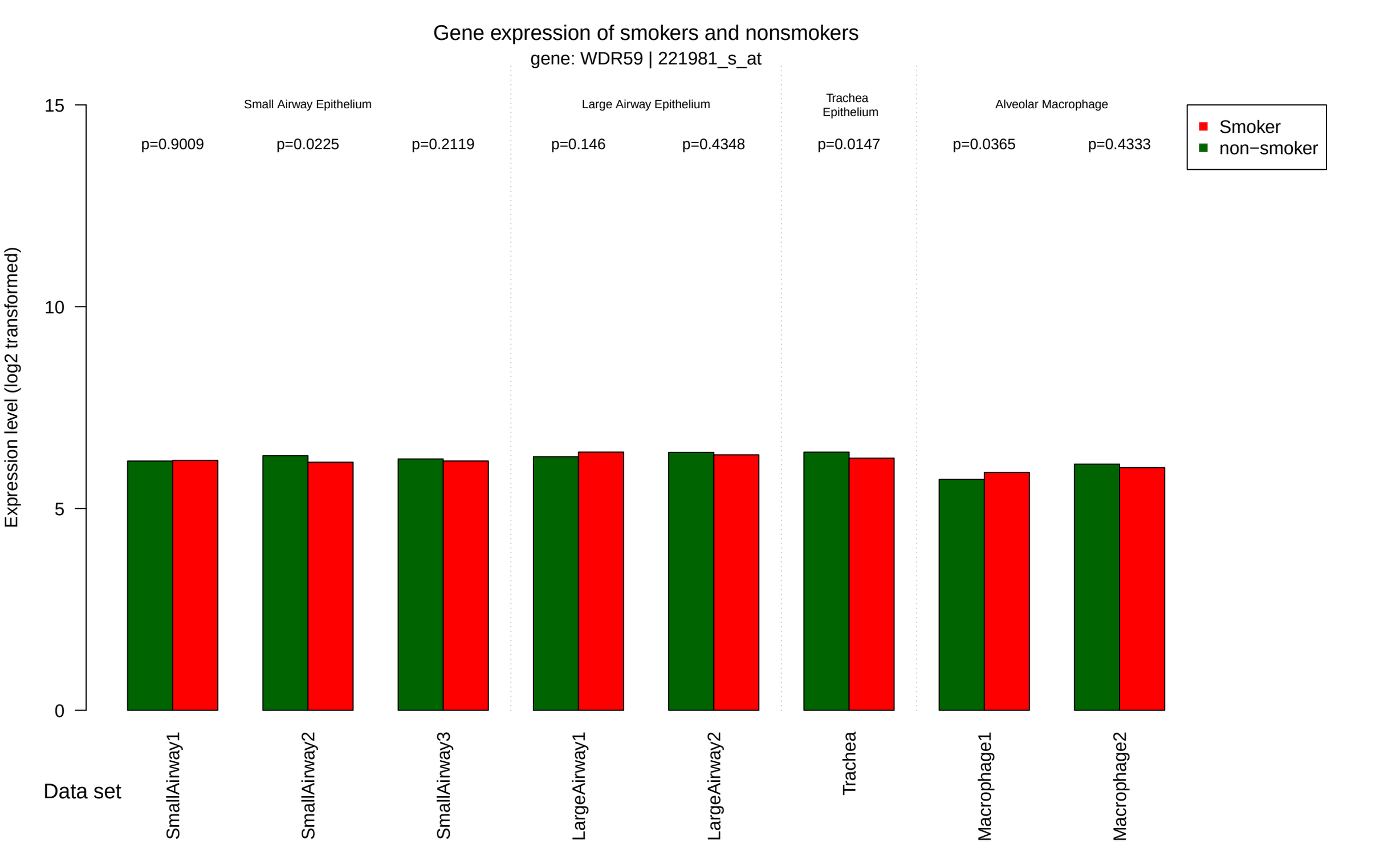
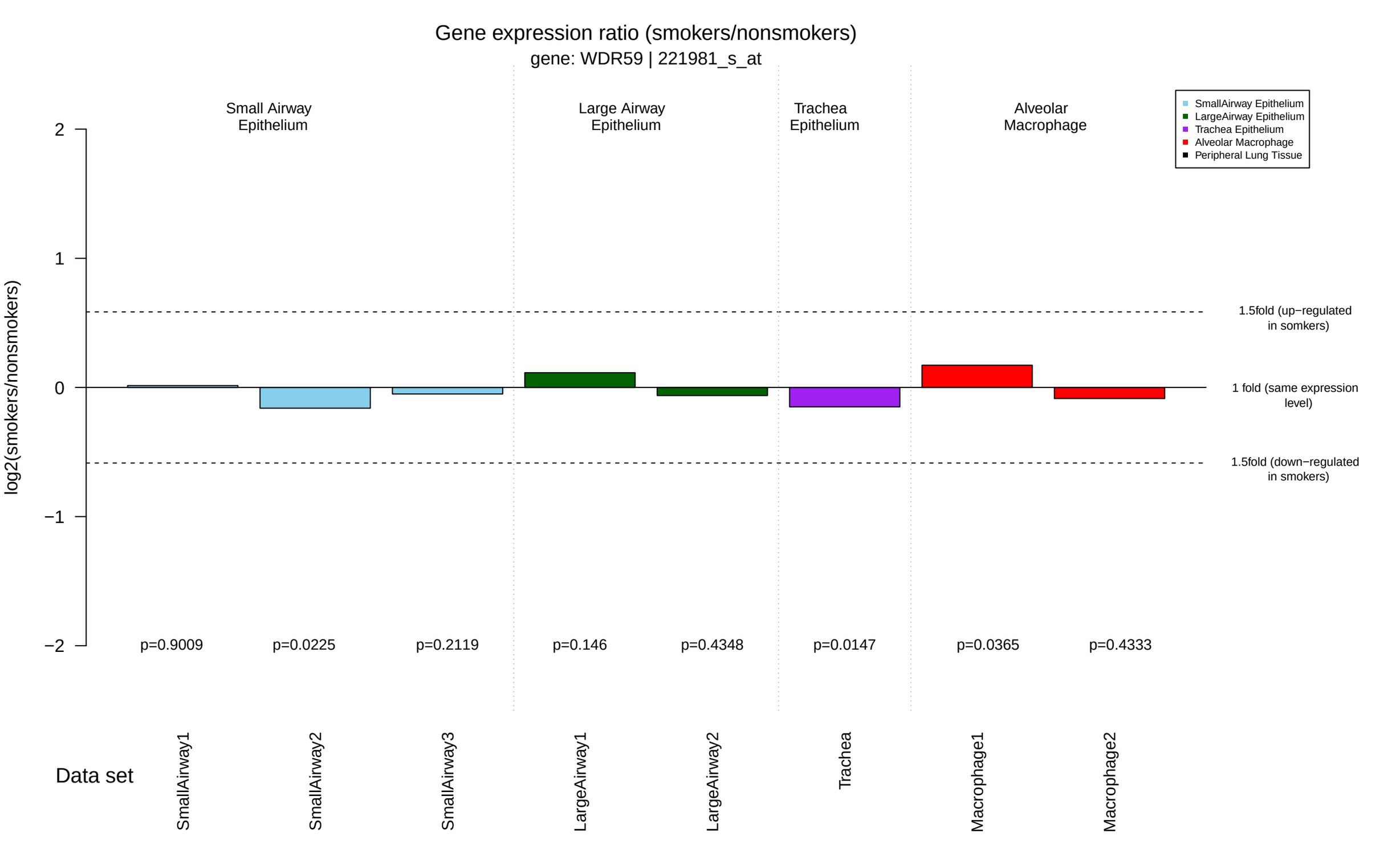
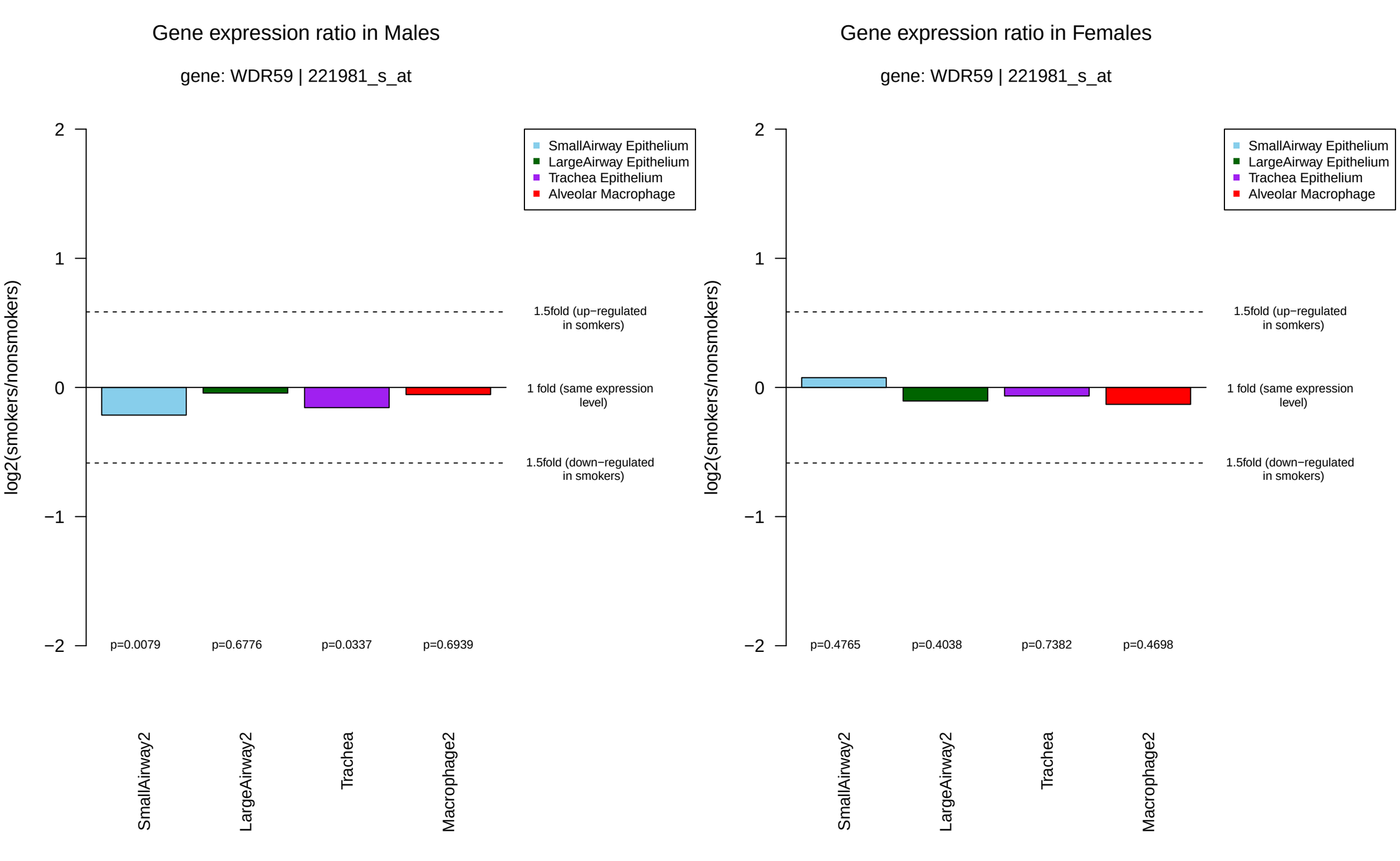
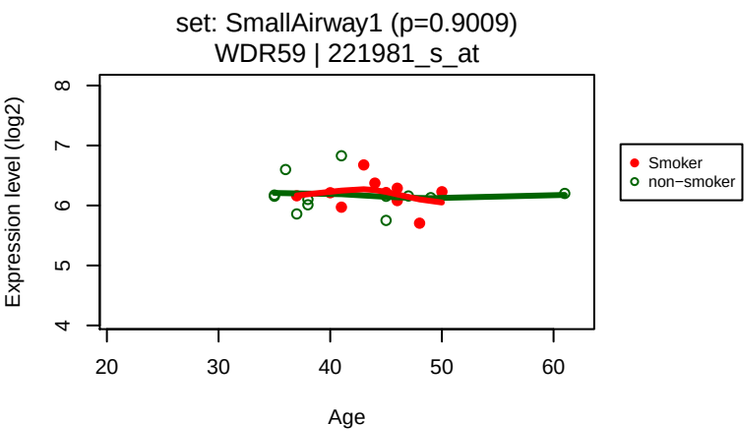
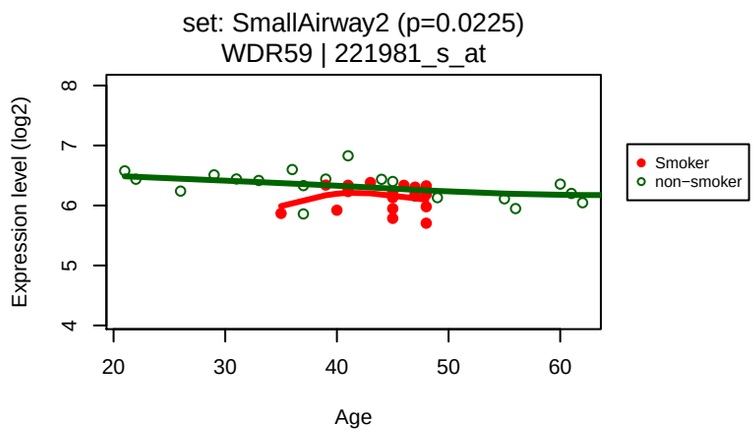
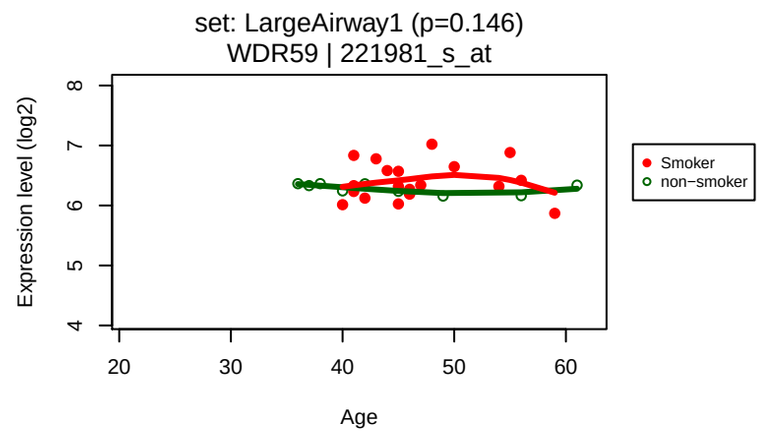

Gene expression of smokers and nonsmokers
gene: WDR59 | 221981_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9009
p=0.0225
p=0.2119
p=0.146
p=0.4348
p=0.0147
p=0.0365
p=0.4333
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

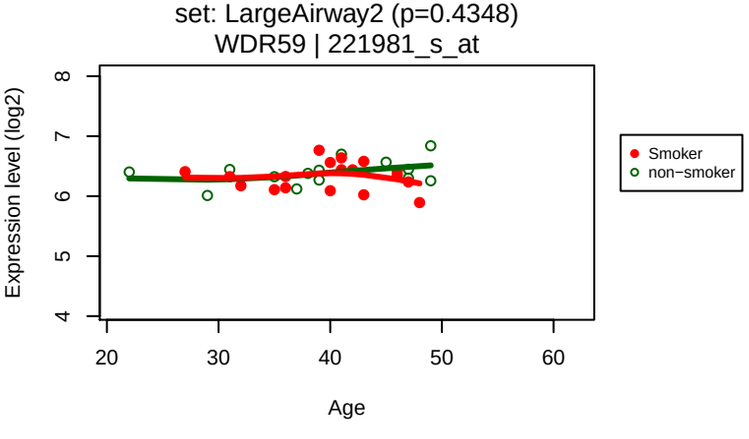

Gene expression ratio (smokers/nonsmokers)
gene: WDR59 | 221981_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9009
p=0.0225
p=0.2119
p=0.146
p=0.4348
p=0.0147
p=0.0365
p=0.4333
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

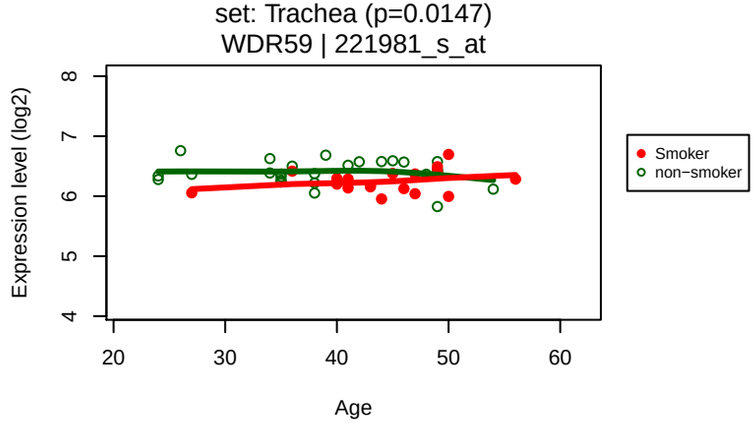

Gene expression ratio in Males
gene: WDR59 | 221981_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0079
p=0.6776
p=0.0337
p=0.6939
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: WDR59 | 221981_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4765
p=0.4038
p=0.7382
p=0.4698
SmallAirway2
LargeAirway2
Trachea
Macrophage2

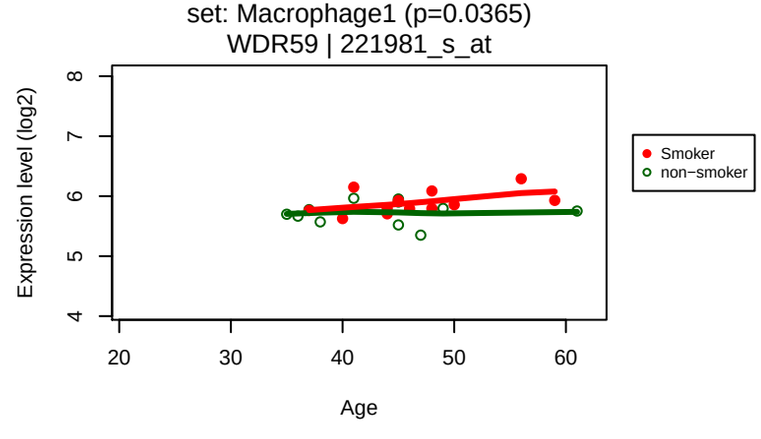

set: SmallAirway1 (p=0.9009)
WDR59 | 221981_s_at
Expression level (log2)
Age
Smoker
non−smoker

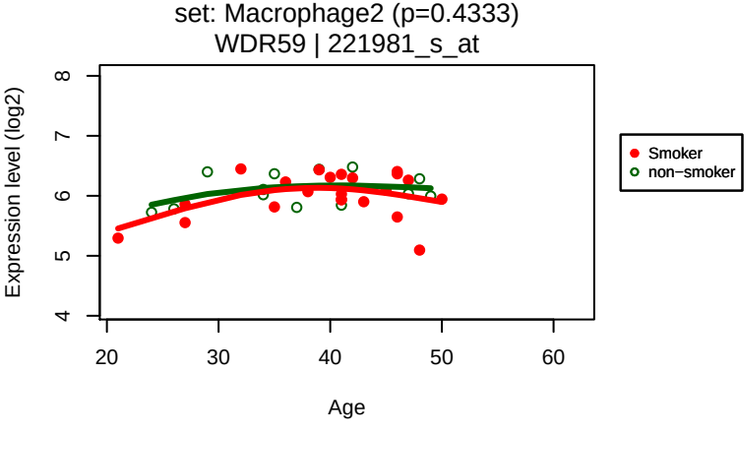

set: SmallAirway2 (p=0.0225)
WDR59 | 221981_s_at
Expression level (log2)
Age
Smoker
non−smoker

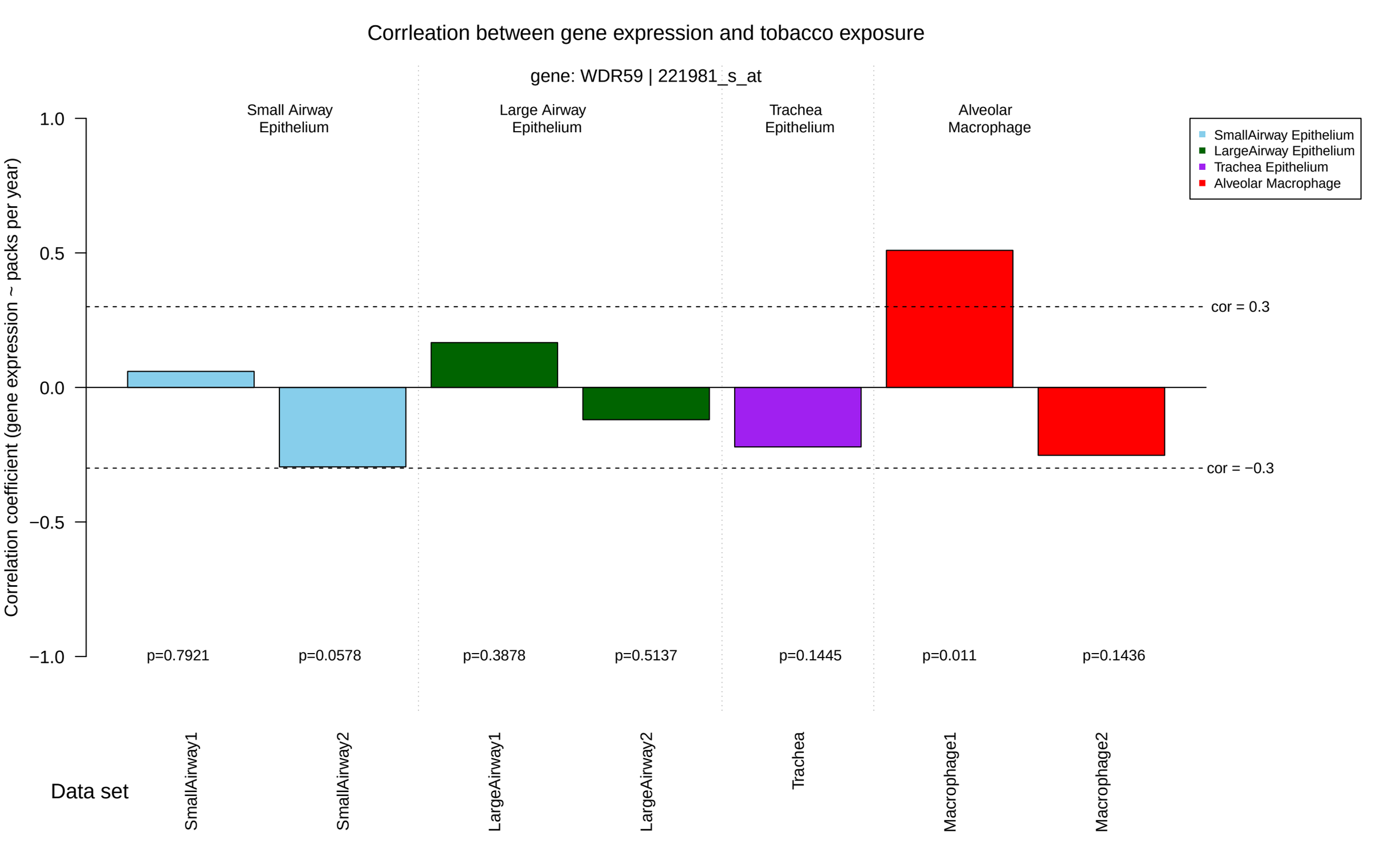

set: LargeAirway1 (p=0.146)
WDR59 | 221981_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4348)
WDR59 | 221981_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0147)
WDR59 | 221981_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0365)
WDR59 | 221981_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4333)
WDR59 | 221981_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: WDR59 | 221981_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7921
p=0.0578
p=0.3878
p=0.5137
p=0.1445
p=0.011
p=0.1436
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2